

IoT-Based Smart Gas Leakage Detection and Emergency Alert System

Cooch Behar Government Engineering College

Electronics and Communication Engineering

Final Year Project



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Introduction

An **IoT-Based Smart Gas Leakage Detection and Emergency Alert System** is a smart safety system designed to detect harmful gas leaks at an early stage. It uses a **gas sensor** to continuously monitor the surrounding air. When a gas leak is detected, the system immediately activates a **buzzer** and sends an **alert notification** to the user's phone through the internet. The system is built using an **ESP32 microcontroller** with built-in **Wi-Fi**, enabling real-time monitoring. This project helps prevent accidents, protects lives and property, and provides a low-cost, reliable, and efficient solution for improving safety in homes, laboratories, and small industries.

Problem Statement

1. Gas Leakage Hazard:

Gas leakage can cause fires, explosions, and serious health risks if not detected on time.

2. Delayed Detection:

Small gas leaks often go unnoticed, increasing the chances of accidents.

3. Limited Traditional Systems:

Conventional gas detectors provide only local alarms and do not support remote monitoring.

4. Lack of Instant Alerts:

Users cannot receive immediate notifications when they are away from the affected location.

5. Need for a Smart IoT Solution:

A low-cost IoT-based system is needed to detect gas leaks in real time and send instant alerts for quick action and improved safety.

Objectives

1. Detect Gas Leakage:

To continuously monitor the environment and detect gas leaks using a gas sensor.

2. Provide Immediate Warning:

To activate a buzzer immediately when a gas leak is detected.

3. Send Real-Time Alerts:

To notify the user instantly through an IoT-based mobile alert system.

4. Enable Remote Monitoring:

To allow users to monitor the gas leakage status from anywhere using the internet.

5. Improve Safety:

To reduce the risk of accidents, protect lives and property, and provide a low-cost, reliable safety solution.

Literature Review

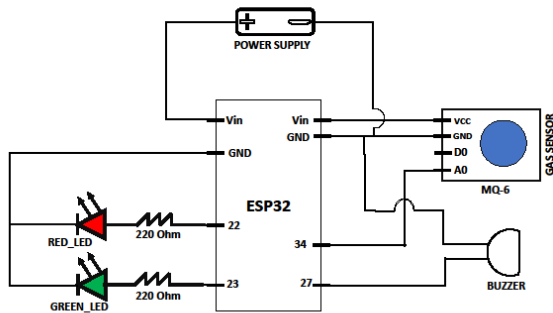
Related Work

- **Traditional Gas Detection Systems** – Used gas sensors and buzzers to detect gas leaks and provide local audio alarms.
- **GSM-Based Gas Leakage Alert Systems** – Sent SMS notifications to users through GSM modules when gas leakage was detected.
- **IoT-Based Gas Monitoring Systems** – Utilized ESP32 with Wi-Fi to monitor gas levels and send real-time alerts via the internet.
- **Cloud-Based Smart Gas Safety Systems** – Enabled remote monitoring, data storage, and notification through cloud platforms and mobile applications.
- **Smart Home Gas Leakage Detection Systems** – Integrated gas sensors with home automation to improve safety through intelligent monitoring and remote access.

Limitations of Existing System

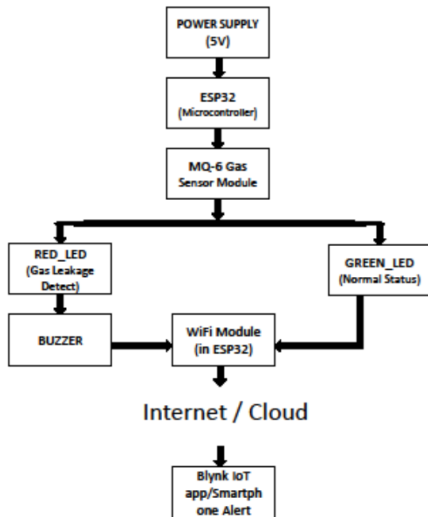
- **Local Alarm Only** – Most traditional systems activate only a buzzer and cannot notify users remotely.
- **No Real-Time Mobile Alerts** – Users do not receive instant notifications when they are away from the location.
- **Internet or Network Dependency** – Many smart systems rely on stable Wi-Fi or GSM networks for communication.
- **High Cost and Complexity** – Advanced gas monitoring systems are expensive and require complex installation and maintenance.
- **Limited Monitoring Features** – Existing systems often focus only on gas detection and lack continuous real-time monitoring and user-friendly remote access.

Circuit Diagram



Smart Gas Leakage Detection and Emergency Alert System

Block Diagram



Working Principle

- **Hardware Development**
 - Assembly of Peltier module with ice mold
 - Integration of heat sink and cooling fan
- **Power and Control**
 - Design and testing of Peltier driver circuit
 - Closed-loop temperature control
- **Software Development**
 - Embedded code for sensing and control
 - Safety and protection features
- **Testing and Optimization**
 - Ice formation time analysis
 - Power consumption measurement
 - Performance improvement
- Final prototype demonstration and documentation

Hardware Components

<u>Component</u>		<u>Purpose</u>
• ESP32	-----	Main microcontroller with built-in Wi-Fi for processing data and sending alerts.
• MQ-6 Gas Sensor	-----	MQ-6 Gas Sensor.
• Buzzer	-----	Produces an audible alarm when gas leakage is detected.
• 3.7V Li-ion Battery	-----	Provides portable power to the system.
• Battery Case	-----	Holds and protects the battery securely.
• Step-Up Boost Converter	-----	Boosts 3.7V battery voltage to 5V for ESP32 and other components.
• Charging Module (TP4056)	-----	Safely charges the Li-ion battery.
• Red LED and Green LED	-----	Indicates gas leakage and safe condition.
• Jumper Wires	-----	Used for electrical connections between components.
• 220 Ω Resistors (2 Nos.)	-----	Limits current for the LEDs.
• Switch	-----	Turns the system ON/OFF.

Software

Software

Purpose

- | | | |
|-------------------------------|-------|---|
| • MIT App Inventor | ----- | Develops the Android mobile application. |
| • Google Firebase | ----- | Stores and synchronizes gas leakage status in real time. |
| • HiveMQ | ----- | MQTT broker for communication between ESP32 and the mobile app. |
| • Notepad | ----- | Used for editing configuration and code files. |
| • Command Prompt (CMD) | ----- | Used for MQTT testing and network-related commands. |

Hardware Setup (Photos)



Smart Gas Leakage Detection and Emergency Alert System



MQ-6 Gas
Sensor



ESP32



Buzzer



Stepup boost
converter



Charging
module



Switch



3.7 v Battery



Led Lights



220 ohm
Resistor (x2)

Results

1. Accurate Gas Leak Detection

The MQ-6 sensor successfully detected LPG gas leakage in real time.

2. Immediate Audio Alert

The buzzer activated instantly when the gas concentration exceeded the predefined threshold.

3. Mobile Notification

The ESP32 successfully sent an emergency alert to the user's mobile application through the IoT platform.

4. Real-Time Monitoring

Gas leakage status was updated and monitored in real time using Google Firebase and HiveMQ.

5. Reliable System Performance

The prototype operated successfully with stable Wi-Fi connectivity and demonstrated fast response, making it suitable for home and small industrial safety applications.

Advantages

1. Early Gas Leak Detection

Detects LPG gas leakage quickly, reducing the risk of accidents.

2. Instant Emergency Alerts

Sends real-time notifications to the user's mobile phone through the Internet.

3. Real-Time Monitoring

Allows users to monitor the gas leakage status remotely using the mobile application.

4. Low Cost and Easy to Build

Uses affordable components like ESP32 and MQ-6, making it suitable for homes and small industries.

5. Reliable and Fast Response

Activates the buzzer immediately and provides quick notification during gas leakage.

6. Portable and Rechargeable

Powered by a rechargeable 3.7V battery, allowing easy installation without a constant power supply.

Future Scope

1. Automatic Gas Supply Shut-Off

Integrate a solenoid valve to automatically stop the gas supply when a leak is detected.

2. Multi-Gas Detection

Upgrade the system to detect multiple gases such as LPG, methane, carbon monoxide, and smoke.

3. GPS-Based Emergency Alerts

Send the exact location of the gas leak to emergency services or family members during critical situations.

4. Low-Power Optimization

Improve battery life using power-saving techniques for long-term operation.

5. Industrial-Scale Deployment

Expand the system for use in factories, laboratories, hotels, and large commercial buildings with multiple gas sensors.

6. Smart Home Integration

Connect the system with smart home platforms like Google Home or Amazon Alexa for voice alerts and automation

Conclusion

The **IoT-Based Smart Gas Leakage Detection and Emergency Alert System** is an effective and affordable solution for improving gas safety. It successfully detects LPG gas leakage using the **MQ-6 sensor**, activates a **buzzer**, and sends **real-time alert notifications** to the user's mobile phone through the Internet. By using **ESP32**, **Google Firebase**, and **HiveMQ**, the system provides reliable remote monitoring and quick emergency response. This project helps reduce the risk of accidents, protects lives and property, and is suitable for homes, laboratories, and small industries.

Thank You